SIROKKO_262.01 — INTEGRATED MINIMAL SAFETY / DRAFT
YD-ESP32-S3 N16R8: native hierarchical schematic candidate.
DRAFT / NOT FOR ORDER / NO POWER / NO FIRE. SCH and PCB are integrated; firmware is not implemented.

THERMOCOUPLES: J30 EXHAUST / J31 AIR — TYPE K
9-pin MAX31856 modules; Adafruit 3263 geometry.
VIN=3V3; 3Vo/FLT=NC; thermocouples at module terminals.
J12 DS18B20 unchanged. J30/J31 and MAX31856 mounting are integrated on the PCB; firmware is not implemented. NO POWER / NO FLASH / NO FIRE.

J30
MAX31856 EXHAUST
TC_EXH_DRDY 1
2
TC_EXH_CS 3
TC_SDI_MOSI 4
TC_SDO_MISO 5
TC_SCK 6
GND 7
8
+3V3 9

J31
MAX31856 AIR
TC_AIR_DRDY 1
2
TC_AIR_CS 3
TC_SDI_MOSI 4
TC_SDO_MISO 5
TC_SCK 6
GND 7
8
+3V3 9

J12
+3V3 1
ROOM_DQ 2
GND 3
ROOM_DS18B20_HARNESS

J1
+3V3 1
+3V3 2
3
ROOM_DQ 4
5
6
7
FUEL_CMD 8
FAN_REQUEST 9
SPEED_CMD 10
11
12
13
14
TC_AIR_CS 15
TC_EXH_CS 16
TC_SDI_MOSI 17
TC_SCK 18
TC_SDO_MISO 19
GLOW_CMD 20
+5V_DEVKIT 21
GND 22
DEVKITC1_J1

J3
GND 1
2
3
TC_EXH_DRDY 4
TC_AIR_DRDY 5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
GND 21
GND 22
DEVKITC1_J3

DRAFT: LM5013 5V is assigned to +5V_DEVKIT.
Installed DevKit power comes only from Sirokko; use USB only with the DevKit removed. NO POWER.

LOGIC_POWER_24_TO_5V__DRAFT_NOT_FOR_ORDER
P24_PROTECTED
GND
P24_PROTECTED
GND
File: logic_power_24_5_DRAFT.kicad_sch

MOTOR_EXTERNAL_HC160
SPEED_CMD
GND
FAN_REQUEST
File: motor_external_hc160.kicad_sch

GLOW_SWITCHED_POS 87
87
K1
30
30 P24_PROTECTED
87a
87a
K_GLOW_COIL_LOW 85
85
86
86 P24_PROTECTED
DG858M-5021-76-1024 + DZ85AB-5-PCB / GLOW

CLUTCH_SWITCHED_POS 87
87
K2
30
30 P24_PROTECTED
87a
87a
K_CLUTCH_RUN_COIL_LOW 85
85
86
86 P24_PROTECTED
DG858M-5021-76-1024 + DZ85AB-5-PCB / CLUTCH_RUN

TWO RELAYS: K1 GLOW; K2 CLUTCH. U11 IN1=GLOW_CMD, IN2=FUEL_CMD; IN3/IN4, ENBL and RESET tied to GND. No independent welded-contact cutoff. Motor control remains external HC160. DRAFT / NO POWER / NO FLASH / NO FIRE.

MAIN INPUT / ROOM SENSOR
J13: 1=VIN_RAW_24, 2=GND. Main design range 21.6..28.8V.
5V conversion is on-board. Installed DevKit power comes only from Sirokko; use USB only with the DevKit removed for programming.
J12: 1=3V3, 2=DQ, 3=GND. Never connect raw 24V to logic terminals.

J13
1 VIN_RAW_24
GND 2 P24_PROTECTED
24V_INPUT_PHOENIX_1756993_WIP
INPUT_PROTECTION_WIP_NOT_QUALIFIED
VIN_RAW_24
P24_PROTECTED
GND
GND
File: Sirokko_Input_Protection_WIP.kicad_sch

+3V3
R1
4k7
ROOM_DQ
+3V3
R2
10k
TC_EXH_CS
+3V3
R3
10k
TC_AIR_CS
FAN_REQUEST
R7
10k
GND
SPEED_CMD
R8
10k
GND

RELAY DRIVER DRAFT / NO POWER / NO FLASH / NO FIRE
U11 DRV8803: GLOW_CMD to IN1; FUEL_CMD to IN2; IN3/IN4, nENBL and RESET to GND.
Exactly K1 glow and K2 clutch; no independent welded-contact cutoff.
Driver, relay-coil, transient and thermal qualification remain open.

REMOTE CONTROL / POWER PATHS — DRAFT
P24 presence monitor removed; J1.6/GPIO6 is NC. Installed DevKit power comes only from Sirokko; use USB only with the DevKit removed for programming.
External watchdog/latch and hardware ARM_PULSE logic have been removed.
Do not restore obsolete permission circuitry. No physical tests performed.

DEFINED INPUTS / OUTPUTS — REMOTE CONTROL ONLY
R1: room-sensor pull-up; R2/R3: thermocouple CS pull-ups.
R7/R8: fan/PWM pulldowns. Relay driver resistors remain inside MinimalSafety.
No display, encoder or Start/Stop button. Their six former DevKit pins are NC.
Remote commands are handled by ESP32; no new control connector is added.
Firmware and remote link-loss behavior remain to be implemented and tested.

MinimalSafety
GND GND
P24_PROTECTED P24_PROTECTED
K_GLOW_COIL_LOW K_GLOW_COIL_LOW
K_CLUTCH_RUN_COIL_LOW K_CLUTCH_RUN_COIL_LOW
GLOW_CMD GLOW_CMD
FUEL_CMD FUEL_CMD
File: MinimalSafety.kicad_sch

J28
CLUTCH_SWITCHED_POS 1
GND 2
CLUTCH_OUTPUT_PHOENIX_1756993

J27 GLOW: 1=switched positive; 2=power GND.
Schematic assigned; PCB current/transient qualification OPEN. NO POWER / NO FIRE.

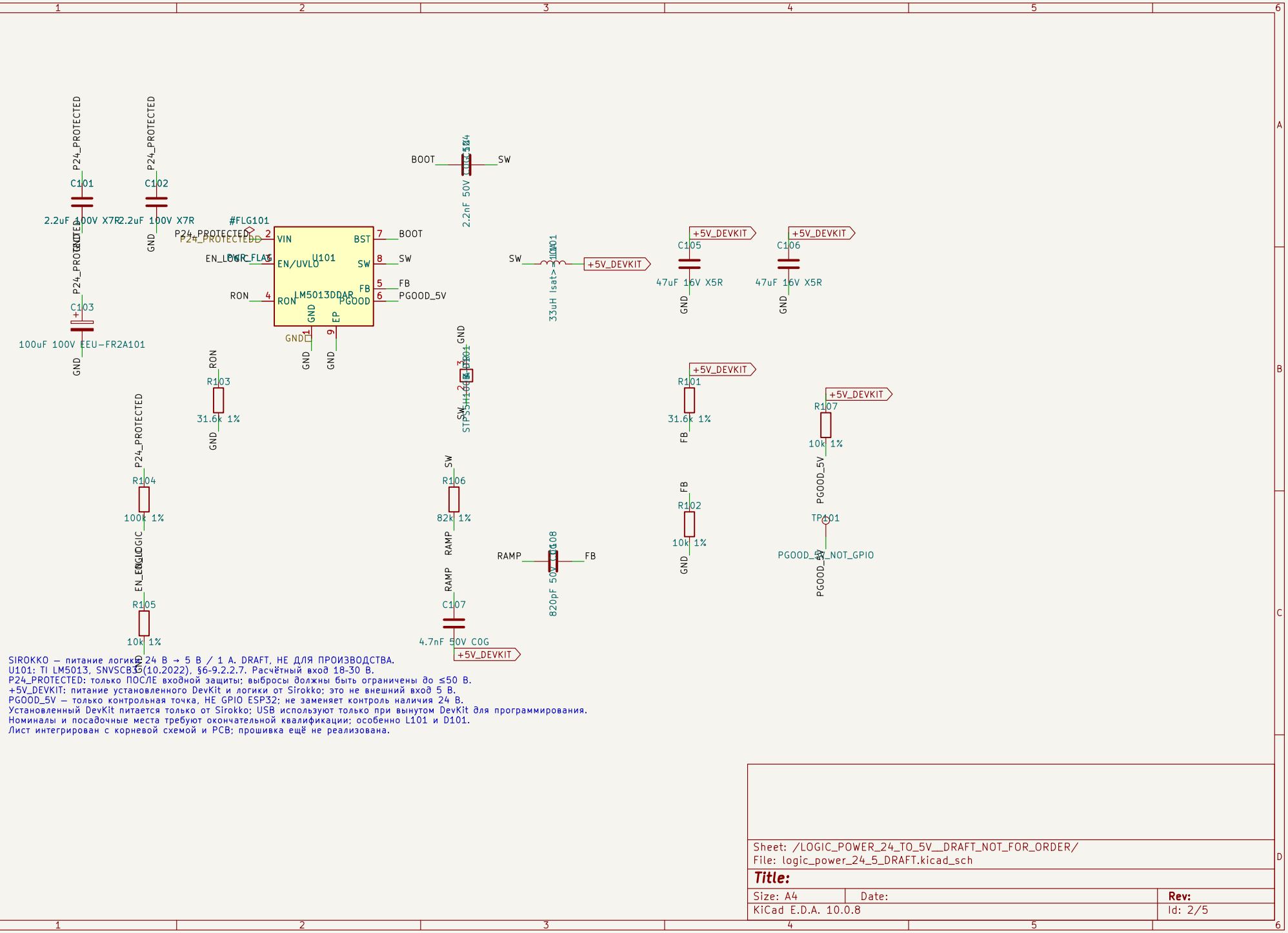
J28 CLUTCH: 1=switched positive; 2=power GND.
Schematic assigned; PCB current/transient qualification OPEN. NO POWER / NO FIRE.

NO LOCAL USER CONTROLS / DRAFT
No display, Start/Stop, encoder or separate RESET on this carrier.
No replacement UI header, jumper or local arming control.
Old external watchdog/latch circuitry is not restored.
DevKit built-in EN/BOOT and USB programming circuitry are unchanged.
No automatic heater restart after reset or a restored remote link.

J27
GLOW_SWITCHED_POS 1
GND 2
GLOW_OUTPUT_PHOENIX_1756993

Interface circuit only; not a burner safety approval

Sheet: /
File: Sirokko262o1_control_motor_candidate.kicad_sch
Title: Sirokko 262.01 — common controller interfaces
Size: A2 | Date: 2026-10-23 | **Rev: DRAFT-01**
KiCad E.D.A. 10.0.8 | Id: 1/5

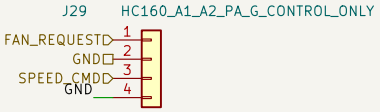


SIROKKO — питание логики 24 В → 5 В / 1 А. DRAFT, НЕ ДЛЯ ПРОИЗВОДСТВА.
U101: TI LM5013, SNVSCB3 (10.2022), §6-9.2.2.7. Расчётный вход 18-30 В.
P24_PROTECTED: только ПОСЛЕ входной защиты; выбросы должны быть ограничены до ≤50 В.
+5V_DEVKIT: питание установленного DevKit и логики от Sirokko; это не внешний вход 5 В.
PGOOD_5V — только контрольная точка, НЕ GPIO ESP32; не заменяет контроль наличия 24 В.
Установленный DevKit питается только от Sirokko; USB используют только при вынутом DevKit для программирования.
Номиналы и посадочные места требуют окончательной квалификации; особенно L101 и D101.
Лист интегрирован с корневой схемой и PCB; прошивка ещё не реализована.

Sheet: /LOGIC_POWER_24_TO_5V__DRAFT_NOT_FOR_ORDER/
File: logic_power_24_5_DRAFT.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 2/5



Sheet: /MOTOR_EXTERNAL_HC160/
File: motor_external_hc160.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 3/5

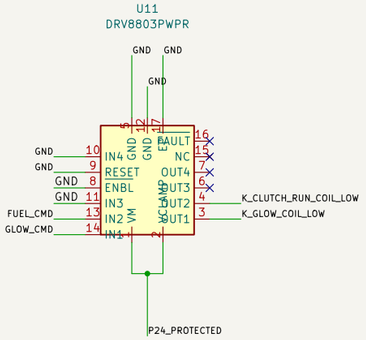
SIROKKO: MINIMAL SAFETY CORE — U11 DRV8803 RELAY DRIVER ONLY

DRAFT / NOT FOR ORDER / NO POWER / NO FIRE — SCH/PCB INTEGRATED; firmware NOT implemented
CURRENT CONTRACT: external TPS3431 watchdog, SN74HCS74 latch and SN74HCS11 safety logic REMOVED / CANCELLED.

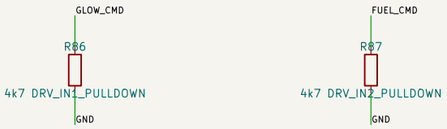
TWO-RELAY DRIVER: IN1=GLOW_CMD; IN2=FUEL_CMD; unused IN3/IN4 and ENBL/RESET=GND. R86/R87 pull commands LOW. OUT3/OUT4 unused. No independent welded-contact cutoff. DRAFT / NO POWER / NO FLASH / NO FIRE.

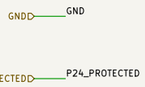
FUTURE FIRMWARE REQUIREMENT (NOT IMPLEMENTED HERE): ESP32-S3 IWDT+TWDT; supervise control task; panic/reboot; keep commands OFF and prohibit automatic restart after reset.

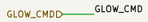
Process checks, timings, purge and software fault handling belong to ESP32; motor/purge power path is unchanged.

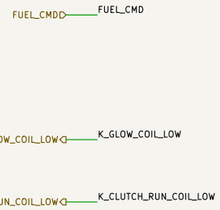






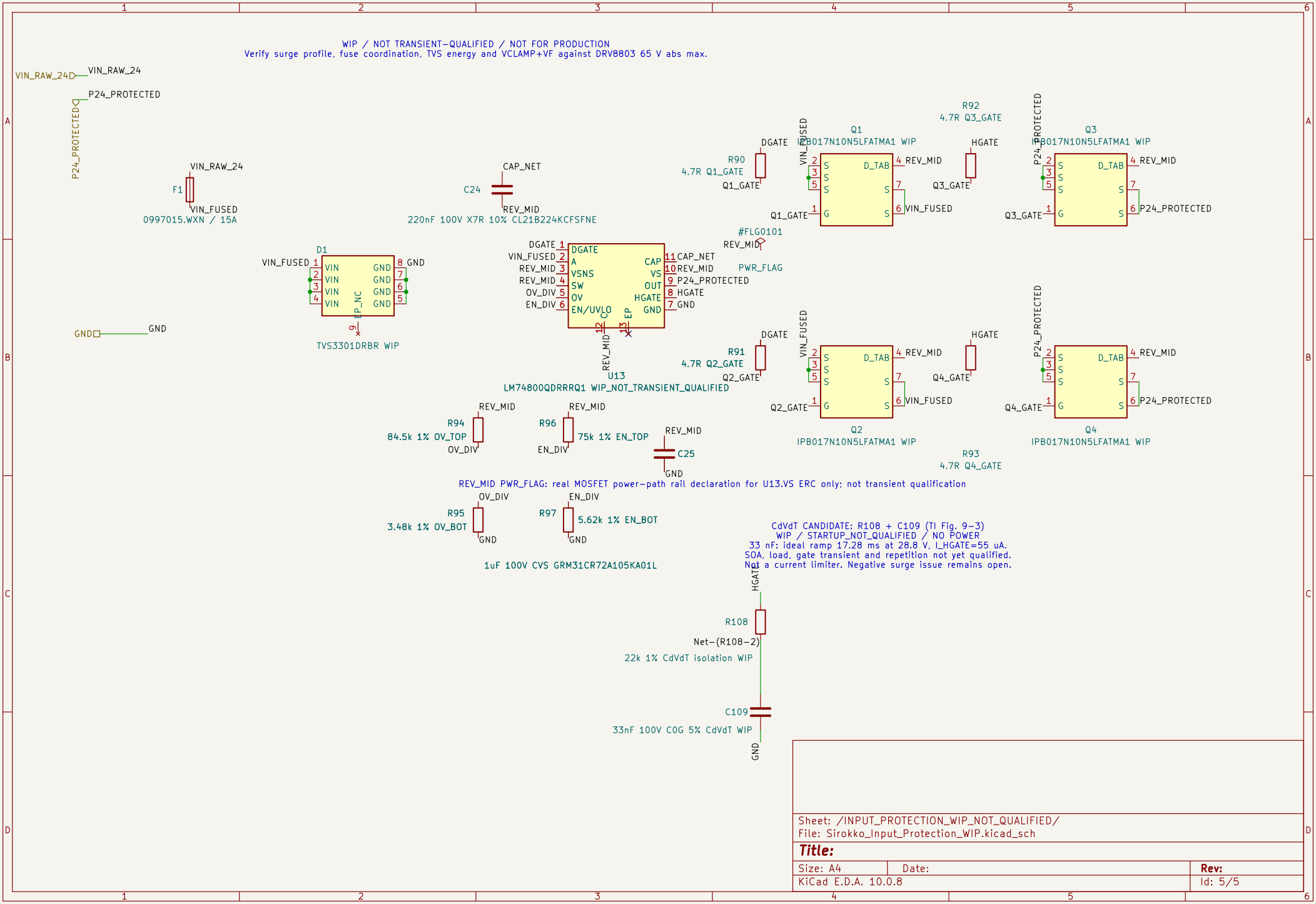






24V loss/recovery, startup dynamics, relay ratings and full safety acceptance remain OPEN.

Sheet: /MinimalSafety/
File: MinimalSafety.kicad_sch
Title:
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.8 | Id: 4/5


WIP / NOT TRANSIENT-QUALIFIED / NOT FOR PRODUCTION
Verify surge profile, fuse coordination, TVS energy and VCLAMP+VF against DRV8803 65 V abs max.
VIN_RAW_24
P24_PROTECTED
F1
0997015.WXN / 15A
C24
220nF 100V X7R 10% CL21B224KCFSFNE
D1
TVS3301DRBR WIP
U13
LM74800QDRRRQ1 WIP_NOT_TRANSIENT_QUALIFIED
#FLG0101
PWR_FLAG
R90
4.7R Q1_GATE
R91
4.7R Q2_GATE
R92
4.7R Q3_GATE
R93
4.7R Q4_GATE
Q1
Q2
Q3
Q4
IPB017N10N5LFATMA1 WIP
R94
84.5k 1% OV_TOP
R95
3.48k 1% OV_BOT
R96
75k 1% EN_TOP
R97
5.62k 1% EN_BOT
C25
REV_MID PWR_FLAG: real MOSFET power-path rail declaration for U13.VS ERC only; not transient qualification
1uF 100V CVS GRM31CR72A105KA01L
CdVdT CANDIDATE: R108 + C109 (TI Fig. 9-3)
WIP / STARTUP_NOT_QUALIFIED / NO POWER
33 nF: ideal ramp 17.28 ms at 28.8 V, I_HGATE=55 uA.
SOA, load, gate transient and repetition not yet qualified.
Not a current limiter. Negative surge issue remains open.
R108
22k 1% CdVdT isolation WIP
C109
33nF 100V C0G 5% CdVdT WIP
Sheet: /INPUT_PROTECTION_WIP_NOT_QUALIFIED/
File: Sirokko_Input_Protection_WIP.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.8
Id: 5/5